A Brief Introduction to Machine Learning

Lumori · Version 1

Learning objectives

1. Define machine learning in terms of algorithms that learn from experience and improve with data [1].

2. Explain why some tasks are better solved by learning from data than by hand-coding explicit rules [1].

3. Describe the standard training process, including parameters and iteration on data [1].

4. Identify the four key components of a machine learning problem: data, model, objective function, and algorithm [1].

5. Distinguish supervised, unsupervised, and reinforcement learning by the kind of feedback or goal involved [1].

6. Explain what deep learning is and how it relates to machine learning and representation learning [1].

What Machine Learning Is

Machine learning is the study of algorithms that can learn from experience. As such an algorithm accumulates more experience, typically in the form of observational data or interactions with an environment, its performance improves. This contrasts with a deterministic e-commerce platform, which follows the same business logic no matter how much experience accrues, until the developers themselves learn and decide it is time to update the software. A summary description adds that machine learning studies how computer systems can leverage experience (often data) to improve performance at specific tasks, and that it combines ideas from statistics, data mining, and optimization; it is often used as a means of implementing AI solutions [1].

Why Learning from Data Matters

Many tasks we would like to automate do not bend easily to human ingenuity. The source lists examples such as predicting tomorrow's weather from geographic information, satellite images, and a trailing window of past weather; answering a factoid question expressed in free-form text; identifying every person in an image and drawing an outline around each; and presenting users with products they are likely to enjoy but unlikely to encounter in the natural course of browsing [1]. The key trick is that even when we do not know how to tell a computer

explicitly how to map from inputs to outputs, we may still be able to perform the cognitive feat ourselves. Even if you do not know how to program a computer to recognize the word 'Alexa', you yourself are able to recognize it. So we can collect a large dataset containing examples of audio snippets and associated labels indicating which snippets contain the wake word [1].

The Training Process and Programming with Data

In the currently dominant approach, we do not attempt to design a system explicitly to recognize wake words. Instead, we define a flexible program whose behavior is determined by a number of parameters, and we use the dataset to determine the best possible parameter values, namely those that improve the performance of our program with respect to a chosen performance measure [1]. If all the knobs are set randomly, the model is unlikely to recognize 'Alexa', 'Apricot', or any other English word. In machine learning, the learning is the process by which we discover the right setting of the knobs for coercing the desired behavior from our model; in other words, we train our model with data. The described training process is: (1) start off with a randomly initialized model that cannot do anything useful; (2) grab some of your data, e.g., audio snippets and corresponding yes/no labels; (3) tweak the knobs to make the model perform better as assessed on those examples; and (4) repeat steps 2 and 3 until the model is awesome [1]. Rather than code up a wake word recognizer, we code up a program that can learn to recognize wake words if presented with a large labeled dataset; this act of determining a program's behavior by presenting it with a dataset can be thought of as programming with data [1].

Key Components of a Machine Learning Problem

In the wake word example, the dataset consists of audio snippets and binary labels. This sort of problem, where we try to predict a designated unknown label based on known inputs given a dataset consisting of examples for which the labels are known, is called supervised learning. The core components that the source says will follow us around no matter what kind of machine learning problem we tackle are: (1) the data that we can learn from; (2) a model of how to transform the data; (3) an objective function that quantifies how well or badly the model is doing; and (4) an algorithm to adjust the model's parameters to optimize the objective function [1].

Kinds of Machine Learning Problems

The wake word problem is just one among many that machine learning can tackle, and the source provides a broad overview of the landscape of machine learning problems [1]. In unsupervised learning, the boss might hand you a giant dump of data and tell you to do some data science with it; this class of problems is called unsupervised learning, and the type and number of questions we can ask is limited only by our creativity. Two example questions are whether we can find a small number of prototypes that accurately summarize the data, such

as grouping photos into landscape photos, pictures of dogs, babies, cats, and mountain peaks, or grouping browsing activities into users with similar behavior (a problem typically known as clustering), and whether we can find a small number of parameters that accurately capture the relevant properties of the data [1]. If you are interested in developing an agent that interacts with an environment and takes actions, you are probably going to focus on reinforcement learning, which may include applications to robotics, dialogue systems, and even AI for video games; deep reinforcement learning applies deep learning to reinforcement learning problems, with prominent examples being the deep Q-network that beat humans at Atari games using only visual input and the AlphaGo program that dethroned the world champion at Go [1].

Deep Learning and the Summary View

Deep learning is one among many popular methods for solving machine learning problems; the source says it will be explained in greater detail later [1]. The summary adds that, as a class of machine learning, representational learning focuses on how to automatically find the appropriate way to represent data. Considered as multi-level representation learning through learning many layers of transformations, deep learning replaces not only the shallow models at the end of traditional machine learning pipelines but also the labor-intensive process of feature engineering. Much of the recent progress in deep learning has been triggered by an abundance of data arising from cheap sensors and Internet-scale applications, and by significant progress in computation, mostly through GPUs. Furthermore, the availability of efficient deep learning frameworks has made design and implementation of whole system optimization significantly easier, and this is described as a key component in obtaining high performance [1].

References

[1] Dive into Deep Learning: Introduction · p. 1

Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola

https://d2l.ai/chapter_introduction/index.html

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